Metal Hanger Manufacturing – Production Process (Software Flow)

1 Raw Material Receipt & Inspection



Process Name in Software: Raw Material Inward (GRN)

Step-by-Step Procedure:

1. Receive steel wire coils from supplier
2. Weigh and verify quantity
3. Check gauge (thickness), tensile strength
4. Perform quality inspection
5. Approve and store in raw material warehouse

Form Fields:

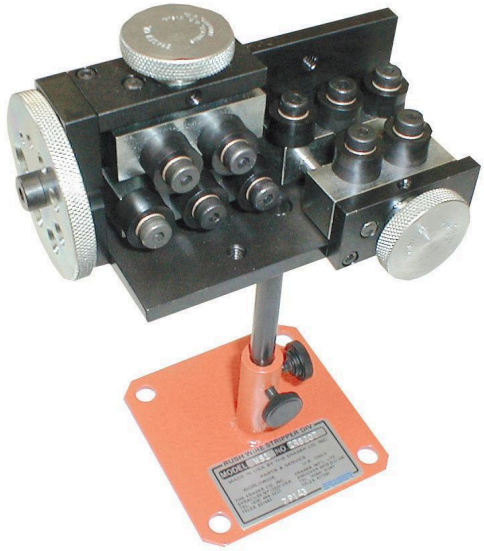
- GRN No (Auto)
- Supplier Name
- Purchase Order No
- Material Name (Steel Wire)
- Wire Gauge (mm)

- Coil Weight (Kg)
- Quantity (No. of Coils)
- Batch/Lot No
- Inspection Status (Pass/Fail)
- Warehouse Location
- Received By
- Date & Time

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2 Wire Straightening & Cutting

Process Name in Software: Wire Processing



Step-by-Step Procedure:

1. Issue raw material to production
2. Load coil into straightening machine
3. Set required hanger length
4. Cut into predefined size pieces
5. Record wastage/scrap

Form Fields:

- Production Order No
- Material Issue No
- Machine ID
- Wire Gauge
- Cutting Length (cm)
- Quantity Planned
- Quantity Produced
- Scrap Quantity
- Operator Name
- Shift
- Start Time / End Time

3️⃣ Hanger Bending & Shaping

Process Name in Software: Hanger Forming

Step-by-Step Procedure:

1. Feed cut wire into bending machine
2. Form shoulder curve
3. Create hook
4. Check dimensions
5. Transfer to finishing

Form Fields:

- Production Order No
- Machine No
- Hanger Type (Adult/Child/Heavy Duty)

- Target Quantity
- Actual Quantity
- Rejected Quantity
- Hook Size (cm)
- Width (cm)
- QC Status
- Operator Name

4️⃣ Surface Treatment / Coating



Process Name in Software: Surface Finishing

Step-by-Step Procedure:

1. Cleaning / Degreasing
2. Rust removal (if required)
3. Powder coating / Painting / Galvanizing
4. Drying / Baking
5. Quality inspection

Form Fields:

- Batch No
- Coating Type (Powder/Galvanized/PVC)
- Color
- Oven Temperature
- Drying Time

- Quantity Processed
- Rejected Quantity
- QC Approval
- Supervisor Name

5️⃣ Final Quality Check

Process Name in Software: Final QC Inspection



Checklist Quality Control

Date 23.12.2023

	High	Medium	Low	Note
Clear Specifications: Ensure that all specifications, standards, and requirements are explicitly defined and understood.	☐	☐	☐	
Raw Materials Inspection: Check incoming materials against quality standards to ensure they meet the necessary criteria.	☐	☐	☐	
Process Control: Monitor and control each stage of the production process to maintain quality standards.	☐	☐	☐	
Equipment Calibration: Regularly calibrate and maintain machinery to guarantee accurate and consistent output.	☐	☐	☐	
Product Testing: Conduct regular testing throughout production to identify defects or deviations from standards.	☐	☐	☐	
Documentation Review: Verify that all documentation, including work instructions and procedures, is up to date and accurate.	☐	☐	☐	
Employee Training: Ensure that all employees involved in the process are adequately trained to maintain quality standards.	☐	☐	☐	
Packaging and Labeling Inspection: Check packaging and labeling to confirm they meet required standards and are accurate.	☐	☐	☐	
Final Product Inspection: Perform a final inspection to confirm the product meets quality standards before it leaves the facility.	☐	☐	☐	

Step-by-Step Procedure:

1. Check shape alignment
2. Measure width & hook size
3. Check coating thickness

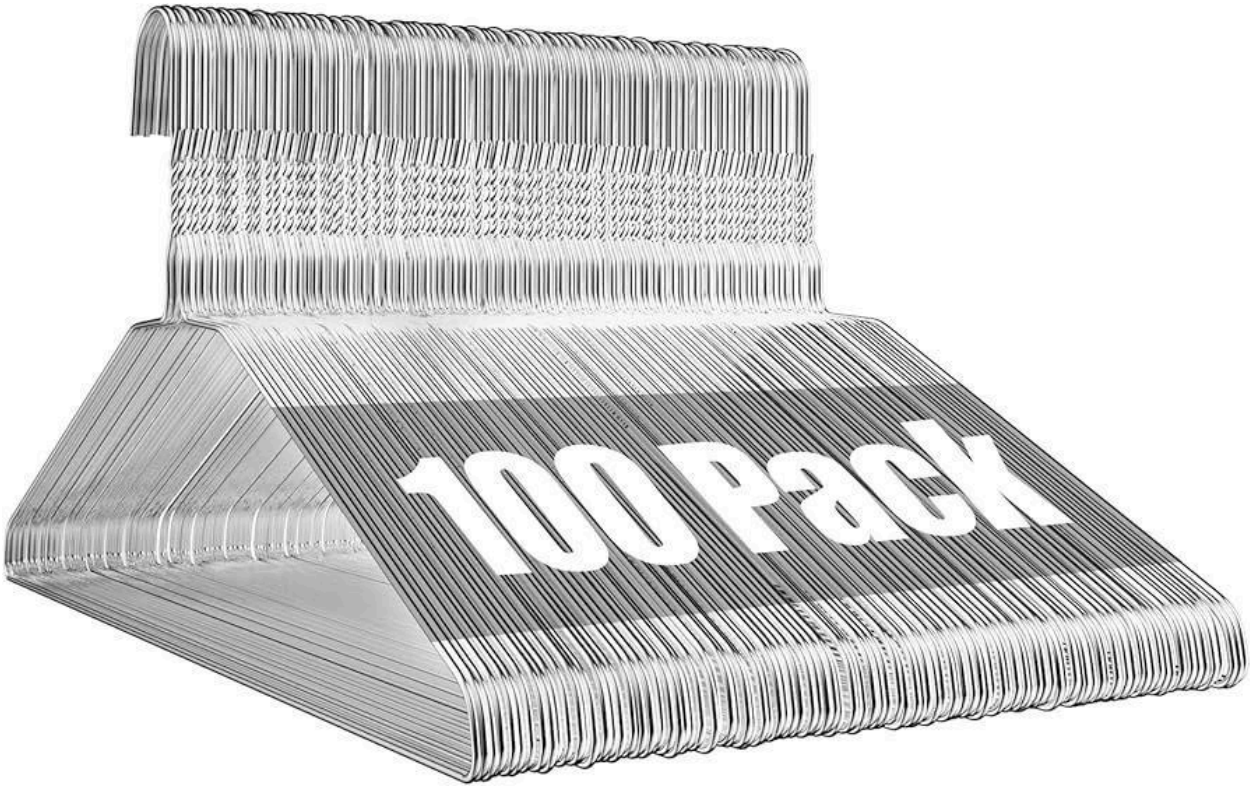
4. Load test (weight bearing)
5. Approve for packing

Form Fields:

- Inspection No
- Production Order No
- Batch No
- Sample Size
- Inspection Checklist (Yes/No)
- Defect Type
- Approval Status
- QC Inspector Name
- Date

6️⃣ Packing & Dispatch

Process Name in Software: Packing & Delivery



Step-by-Step Procedure:

1. Count approved hangers
2. Bundle (e.g., 10/50/100 pcs)
3. Pack in cartons
4. Generate packing list
5. Dispatch to customer

Form Fields:

- Packing Batch No
- Customer Name
- Sales Order No
- Hanger Type
- Quantity per Bundle

- Total Bundles
- Total Quantity
- Carton No
- Delivery Note No
- Dispatch Date
- Transport Details

📊 **Complete ERP Production Flow (Module Structure)**

1. Sales Order
2. Production Planning
3. Raw Material Issue
4. Wire Processing
5. Forming
6. Coating
7. QC
8. Packing
9. Inventory Update (Finished Goods)
10. Delivery & Invoice

Metal Hanger Production – Flowchart (Coil → Batch Manufacturing)

Stage 1: Raw Material (Coil Based Production)







High-Level Flow (Coil to Finished Goods)

START
↓
Receive Steel Wire Coil (GRN Entry)
↓
Quality Inspection (Gauge / Weight Check)
↓
Store in Raw Material Warehouse
↓
Create Production Order
↓
Issue Coil to Production
↓
Wire Straightening & Cutting
↓
Batch Creation (Based on Hanger Type)
↓
Bending & Hook Forming
↓
Surface Coating / Galvanizing

↓

Final Quality Check

↓

Packing (Bundle & Carton)

↓

Finished Goods Inventory

↓

Dispatch

END

Stage 2: Batch Manufacturing Concept

In hanger production, one **coil** produces multiple **batches**.

Example:

1 Coil = 500 Kg

1 Hanger Weight = 50g

Total Possible Hangers = 10,000 pcs

Batches Created:

Batch 1 – 3,000 pcs (Adult White)

Batch 2 – 2,000 pcs (Child Blue)

Batch 3 – 5,000 pcs (Heavy Duty Black)

Detailed Batch-Based Flowchart

Coil Inward

GRN No

Coil No

Weight (Kg)

Gauge (mm)

Supplier

↓

2 Production Order Creation

Production Order No

Hanger Type

Planned Quantity

Required Wire (Auto Calculation)

Target Batch Size

↓

Material Issue (Partial Issue Allowed)

Coil No

Issued Weight

Balance Weight (Auto Update)

↓

Batch Creation

Each batch will have:

Batch No (Auto Generated)

Production Order Ref

Hanger Type

Color

Planned Qty

Shift

Machine No

↓

Wire Processing (Per Batch)

Cutting Length

Quantity Produced

Scrap Qty

Operator

11. ↓

12. 6 Forming Process

13. Shoulder Width

14. Hook Size

15. Rejected Qty

16. ↓

17. 7 Coating / Finishing

18. 





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19. Coating Type
20. Color
21. Oven Temperature
22. Batch Qty
23. ↓

Final QC (Batch Wise Approval)

24. Sample Size
25. Defect %
26. Approval Status
27. QC Signature
28. ↓

Packing (Batch Wise)

29. Bundle Size (10/50/100)
30. Total Bundles

31. Carton No
32. Packing Date
33. ↓

Finished Goods Entry

34. Batch No
35. Quantity
36. Warehouse Location
37. Cost per Unit (Auto from Production Costing)

ERP Software Flow Structure (Database Level)

38. Tables Structure Suggestion

39. raw_material_coil
40. production_order
41. material_issue
42. production_batch
43. batch_processing
44. batch_qc
45. batch_packing
46. finished_goods_stock

Smart ERP Controls You Should Add

47. ✅ Auto wire consumption calculation
✅ Scrap percentage tracking
✅ Machine-wise productivity
✅ Shift-wise production report
✅ Coil balance tracking
✅ Batch traceability (Coil → Batch → Customer)
✅ Cost per hanger auto calculation